

Trigonometry

CHAPTER 6

Height and Distance

6.1–6.4

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Height and Distance

INTRODUCTION

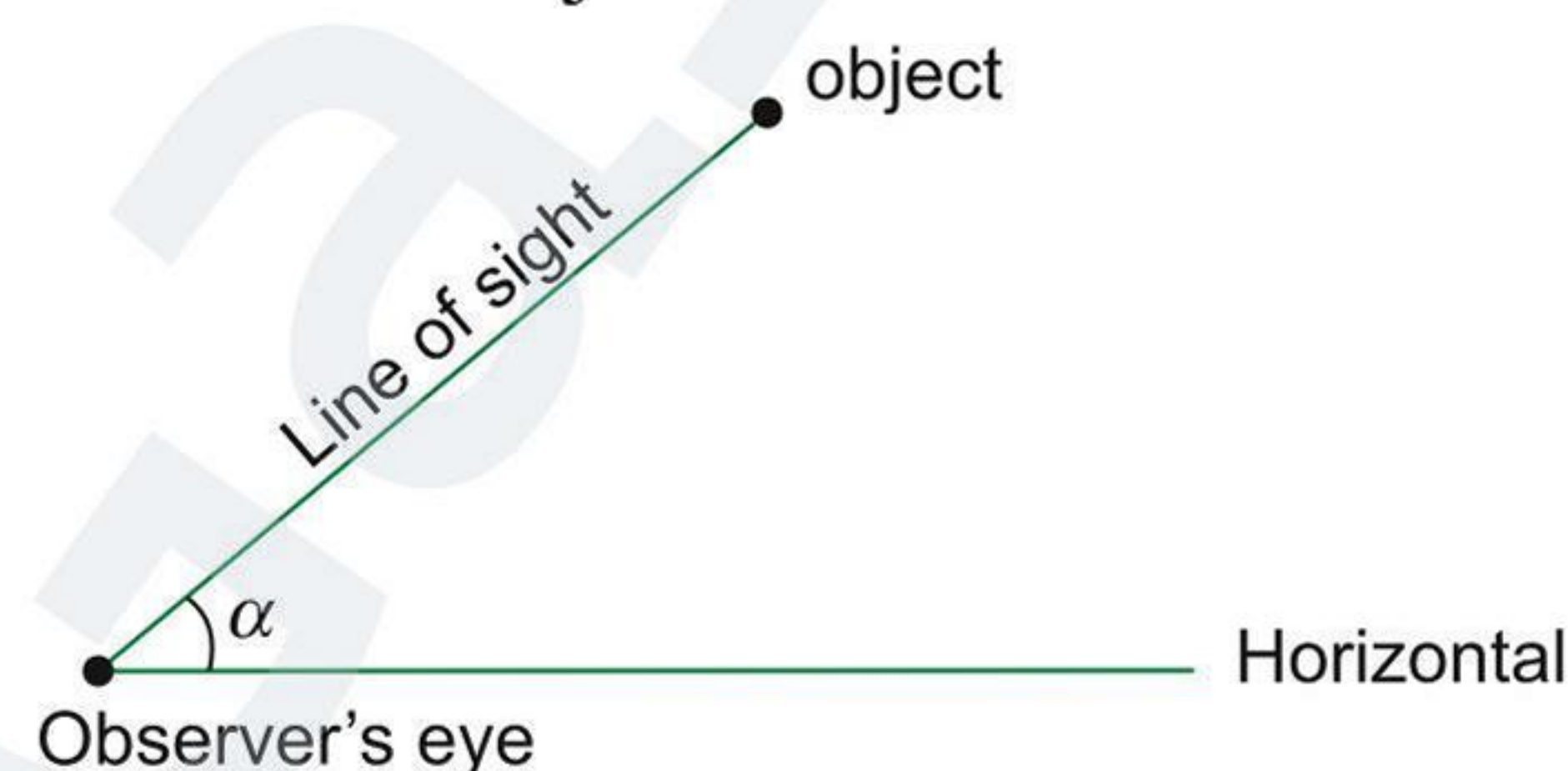
We have studied about trigonometric ratios and their applications in solving triangles involved in different cases. These ratios are also useful to solve problems regarding heights and distances around us in real life. This is why scholars around the world have been studying trigonometry for ages. It is not possible to measure every distance using measuring tape, for instance, altitude of an aeroplane at a certain time, distance of a ship from a light house, height of a hill (distance between its foot and summit), distance between two celestial objects etc.. To measure such distances, scientists developed the method of trigonometric ratios. When dealing with heights and depths, we have to measure two types of angles (above and below the observer's eye-level). The instruments called theodolite and sextant are used to measure these angles and then the method of solution of triangles is used to find the required height or distance.

ANGLE OF ELEVATION AND DEPRESSION

ANGLE OF ELEVATION

If the object is above the observer's eye-level, the angle between the horizontal line and the line of sight (line from the observer's

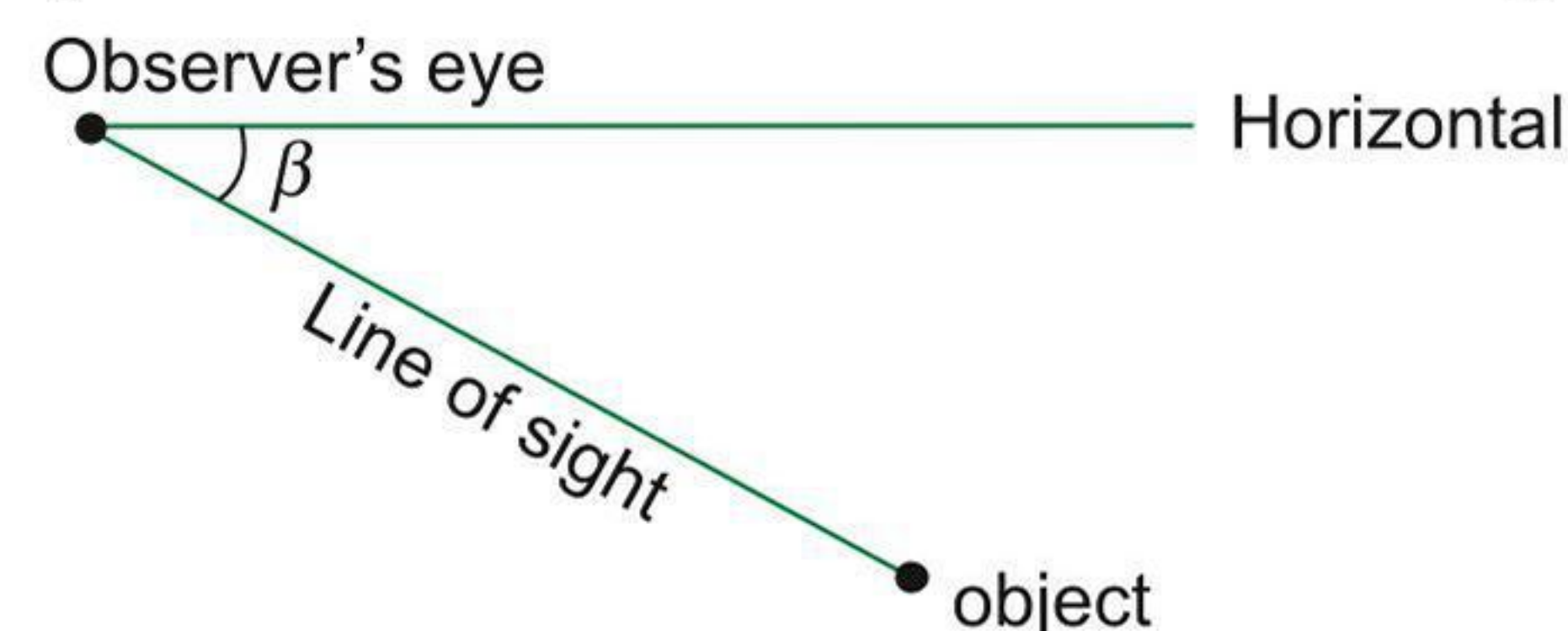
eye to the object) is called the angle of elevation of the object. Here, the horizontal line is taken in the same vertical plane as that of the observer and the object.



In the figure, α is the angle of elevation.

ANGLE OF DEPRESSION

If the object is below the observer's eye-level, then the angle between the horizontal and the observer's line of sight is called the angle of depression. Here, the horizontal line is taken in the same vertical plane as that of the observer and the object.



In the figure, β is the angle of depression.

Exercises

Single Correct Answer Type

- The tops of two poles of height 20 m and 14 m are connected by a wire. If the wire makes an angle 30° with the horizontal, then the length of the wire is
(1) 8 m (2) 12 m (3) 10 m (4) 3 m
- The angle of elevation of the top of an unfinished tower at a point distant 120 m from its base is 45° . If the elevation of the proposed top at the same point is to be 60° , the tower must be raised to a height
(1) $120(\sqrt{3} + 1)$ m (2) $120(\sqrt{3} - 1)$ m
(3) $120\sqrt{3}$ m (4) 120 m
- A tower of height b subtends an angle at a point O on the ground level through the foot of the tower and at a distance a from the foot of the tower. A pole mounted on the top of the tower also subtends an equal angle at O . The height of the pole is
(1) $a\left(\frac{a^2 - b^2}{a^2 + b^2}\right)$ (2) $a\left(\frac{a^2 + b^2}{a^2 - b^2}\right)$
(3) $b\left(\frac{a^2 - b^2}{a^2 + b^2}\right)$ (4) $b\left(\frac{a^2 + b^2}{a^2 - b^2}\right)$
- A ladder rests against a wall making an angle α with the horizontal. The foot of the ladder is pulled away from the wall through a distance x , so that it slides a distance y down the wall making an angle β with the horizontal. The correct relation is
(1) $y = x \tan \frac{\alpha + \beta}{2}$ (2) $x = y \tan \frac{\alpha + \beta}{2}$
(3) $x = y \tan(\alpha + \beta)$ (4) $y = x \tan(\alpha + \beta)$
- Two flagstaffs stand on a horizontal plane. A and B are two points on the line joining their feet and between them. The angles of elevation of the tops of the flagstaffs as seen from A are 30° and 60° and as seen from B are 60° and 45° . If AB is 30 m, then the distance between the flagstaffs is
(1) $30 + 15\sqrt{3}$ (2) $45 + 15\sqrt{3}$
(3) $60 - 15\sqrt{3}$ (4) $60 + 15\sqrt{3}$
- A snake observes an eagle perching on the top of a pole 20 m high. Its elevation from snake's eye is 45° before it flies off horizontally straight away from the snake and after one second the elevation of the eagle reduces to 30° . The speed of the eagle is
(1) 14.64 m/s (2) 17.71 m/s
(3) 12 m/s (4) None of these
- For a man, the angle of elevation of the highest point of a tower situated west to him is 60° . On walking 240 metres to north, the angle of elevation reduces to 30° . The height of the tower is
(1) $50\sqrt{3}$ m (2) $30\sqrt{6}$ m
(3) $60\sqrt{6}$ m (4) 60 m
- A flagstaff stands in the centre of a rectangular field whose diagonal is 120 m. It subtends angles of 15° and 45° at the midpoints of the sides of the field. The height of the flagstaff is
(1) 20 m (2) $30\sqrt{2 + \sqrt{3}}$ m
(3) $30\sqrt{2 - \sqrt{3}}$ m (4) 40 m
- AB is a vertical pole resting at the end A on the level ground. P is a point on the level ground such that $AP = 3AB$ and C is the midpoint of AB . If AC and CB subtend angles α and β , respectively, at P , then the value of $\tan \beta$ is
(1) $18/19$ (2) $3/19$ (3) $1/6$ (4) $1/3$
- From the bottom of a pole of height h , the angle of elevation of the top of a tower is α and the pole subtends angle β at the top of the tower. The height of the tower is
(1) $\frac{h \cot(\alpha - \beta)}{\cot(\alpha - \beta) - \cot \alpha}$ (2) $\frac{h \tan(\alpha - \beta)}{\tan(\alpha - \beta) - \tan \alpha}$
(3) $\frac{\cot(\alpha - \beta)}{\cot(\alpha - \beta) - \cot \alpha}$ (4) None of these
- A tower subtends an angle α at a point on the same level as the foot of the tower. At a second point, b meters above the first, the angle of depression of the foot of the tower is β . The height of the tower is
(1) $b \cot \alpha \tan \beta$ (2) $b \tan \alpha \tan \beta$
(3) $b \tan \alpha \cot \beta$ (4) $b \cot \alpha \cot \beta$
- A man standing on a level plane observes the elevation of the top of a pole to be θ . He then walks a distance equal to double the height of the pole and then finds that the elevation is now 2θ . The value of $\cot \theta$ is
(1) $\sqrt{2} + 1$ (2) $2 - \sqrt{3}$
(3) $\sqrt{2} - 1$ (4) $2 + \sqrt{3}$
- A 5 m high pole stands on a building of height 25 m. The pole and the building subtend equal angles at an antenna placed at a height of 30 m. The distance of the antenna from the top of the pole is
(1) $5\sqrt{\frac{2}{3}}$ (2) $\frac{5\sqrt{3}}{2}$
(3) $5\sqrt{\frac{3}{2}}$ (4) $5\sqrt{6}$
- A vertical tower stands on a declivity which is inclined at 15° to the horizontal. From the foot of the tower, a man ascends the declivity for 80 feet and then finds that the tower subtends an angle of 30° . The height of the tower is
(1) $40(\sqrt{6} + \sqrt{2})$ (2) $20(\sqrt{6} - \sqrt{2})$
(3) $40(\sqrt{6} - \sqrt{2})$ (4) $80(\sqrt{6} - \sqrt{2})$
- The length of the shadow of a pole inclined at 10° to the vertical towards the sun is 2.05 metres, when the elevation of the sun is 38° . The length of the pole is

- (1) $\frac{2.05 \sin 42^\circ}{\sin 38^\circ}$ (2) $\frac{2.05 \cos 38^\circ}{\cos 42^\circ}$
 (3) $\frac{2.05 \sin 38^\circ}{\sin 42^\circ}$ (4) None of these

16. A tower subtends angles α , 2α , and 3α , respectively, at points A , B and C all lying on a horizontal line through the foot of the tower. Then $\frac{AB}{BC} =$

- (1) $\frac{3 \sin \alpha}{\sin 2\alpha}$ (2) $1 + 2 \cos 2\alpha$
 (3) $2 + \cos 3\alpha$ (4) $\frac{\sin 2\alpha}{\sin \alpha}$

17. A harbour lies in a direction 60° south-west from a fort and at a distance 30 km from it. A ship sets out from the harbour at noon and sails due east at 10 km/hour. The ship will be 70 km from the fort at

- (1) 7 p.m. (2) 8 p.m. (3) 5 p.m. (4) 10 p.m.

18. A tower AB leans towards west making an angle α with the vertical. The angular elevation of B , the topmost point of the tower, is β , as observed from a point C due east of A at a distance d from A . If the angular elevation of B from a point D at a distance $2d$ due east of C is γ , then

- (1) $2 \tan \alpha = 2 \cot \beta - \cot \gamma$
 (2) $2 \tan \alpha = 3 \cot \beta - \cot \gamma$
 (3) $\tan \alpha = \cot \beta - \cot \gamma$
 (4) none of these

Answers Key

EXERCISES

Single Correct Answer Type

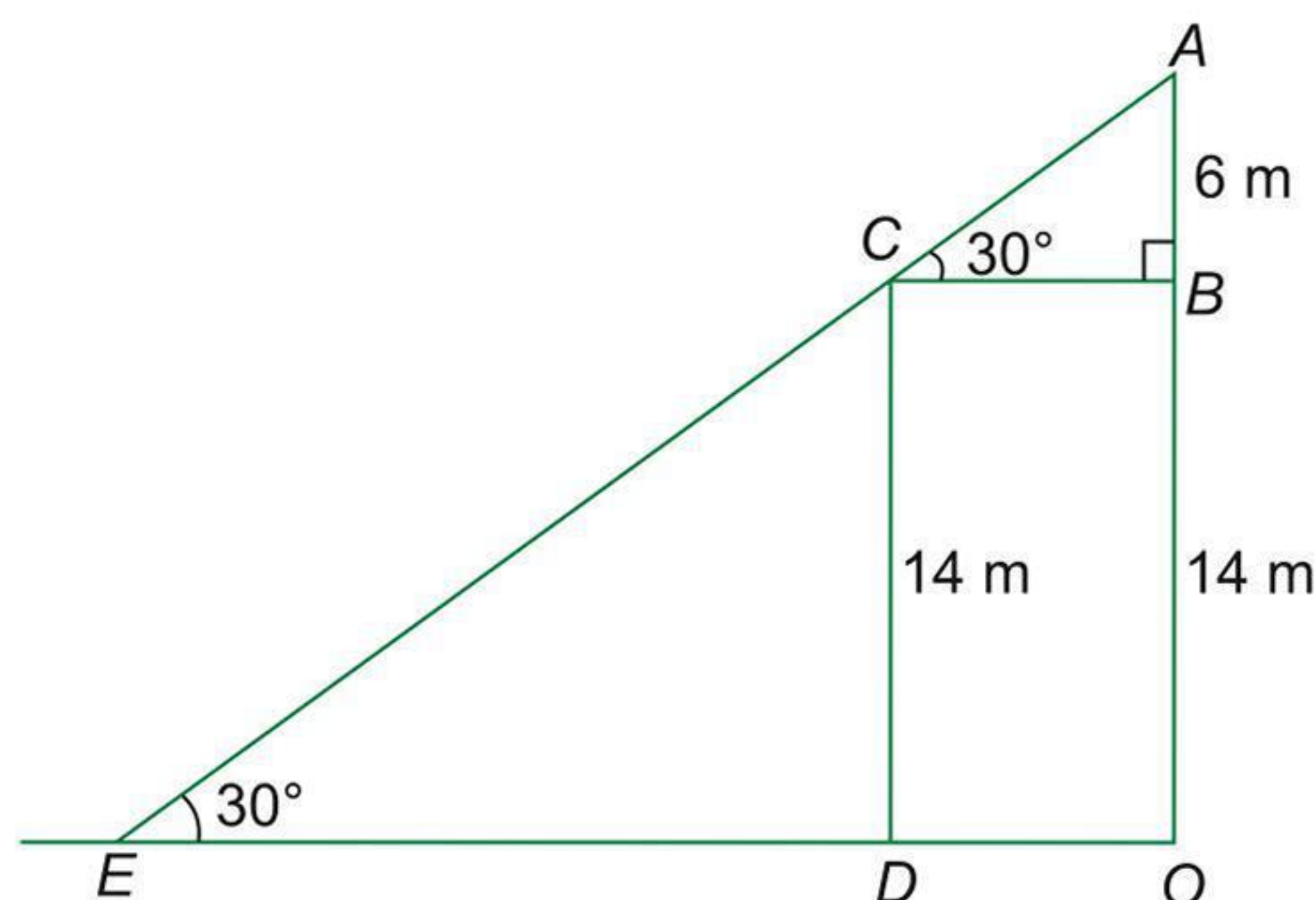
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|--------|--------|--------|--------|---------|---------|---------|---------|---------|---------|
| 1. (2) | 2. (2) | 3. (4) | 4. (2) | 5. (4) | 11. (3) | 12. (4) | 13. (3) | 14. (3) | 15. (3) |
| 6. (1) | 7. (3) | 8. (3) | 9. (2) | 10. (1) | 16. (2) | 17. (2) | 18. (2) | | |

Chapter 6

Exercises

Single Correct Answer Type

1. (2) Let AO and CD be the towers.



In triangle ABC ,

$$AC = 6 \operatorname{cosec} 30^\circ = 12 \text{ m}$$

2. (2) Let OA be the unfinished tower and it is to be raised to point B . Also, let C be the point of observation.

In the figure,

$$\frac{h+x}{120} = \tan 60^\circ = \sqrt{3}$$

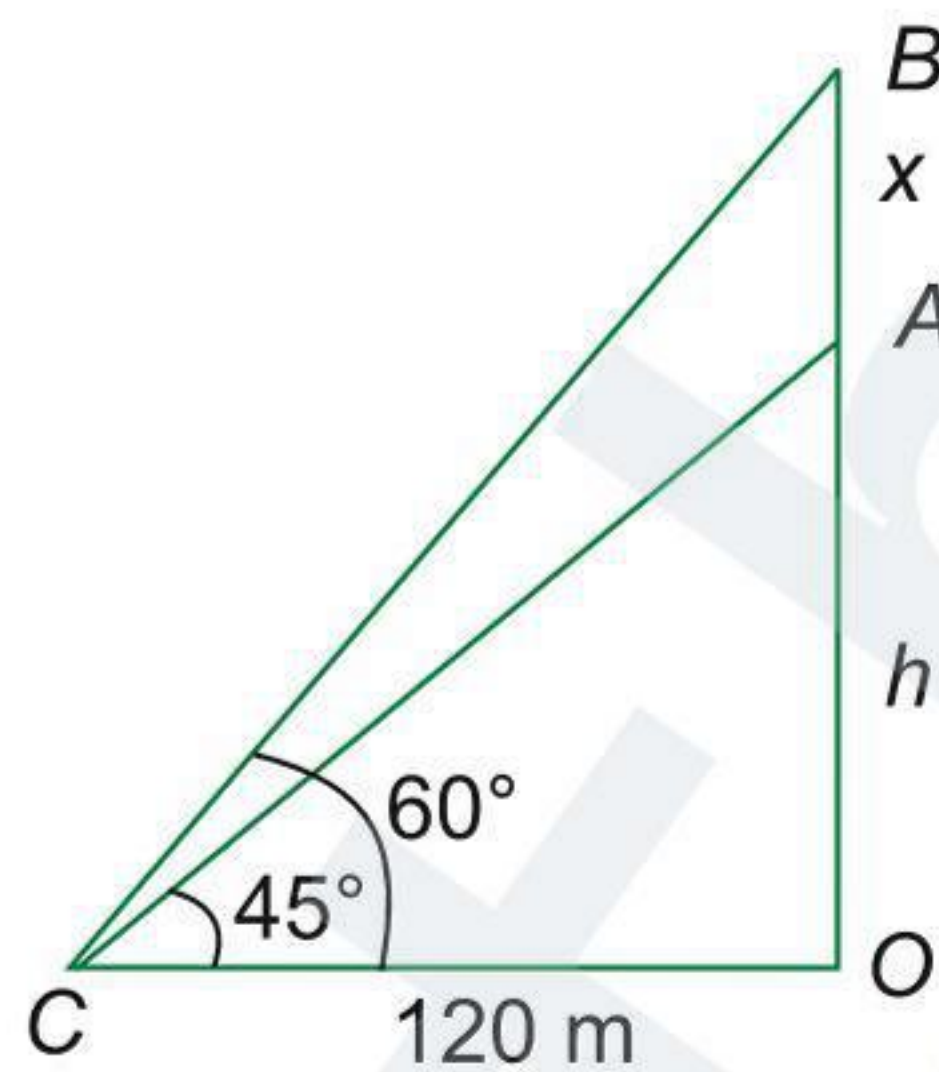
$$\therefore h+x = 120\sqrt{3}$$

$$\text{Also, } \frac{h}{120} = \tan 45^\circ = 1$$

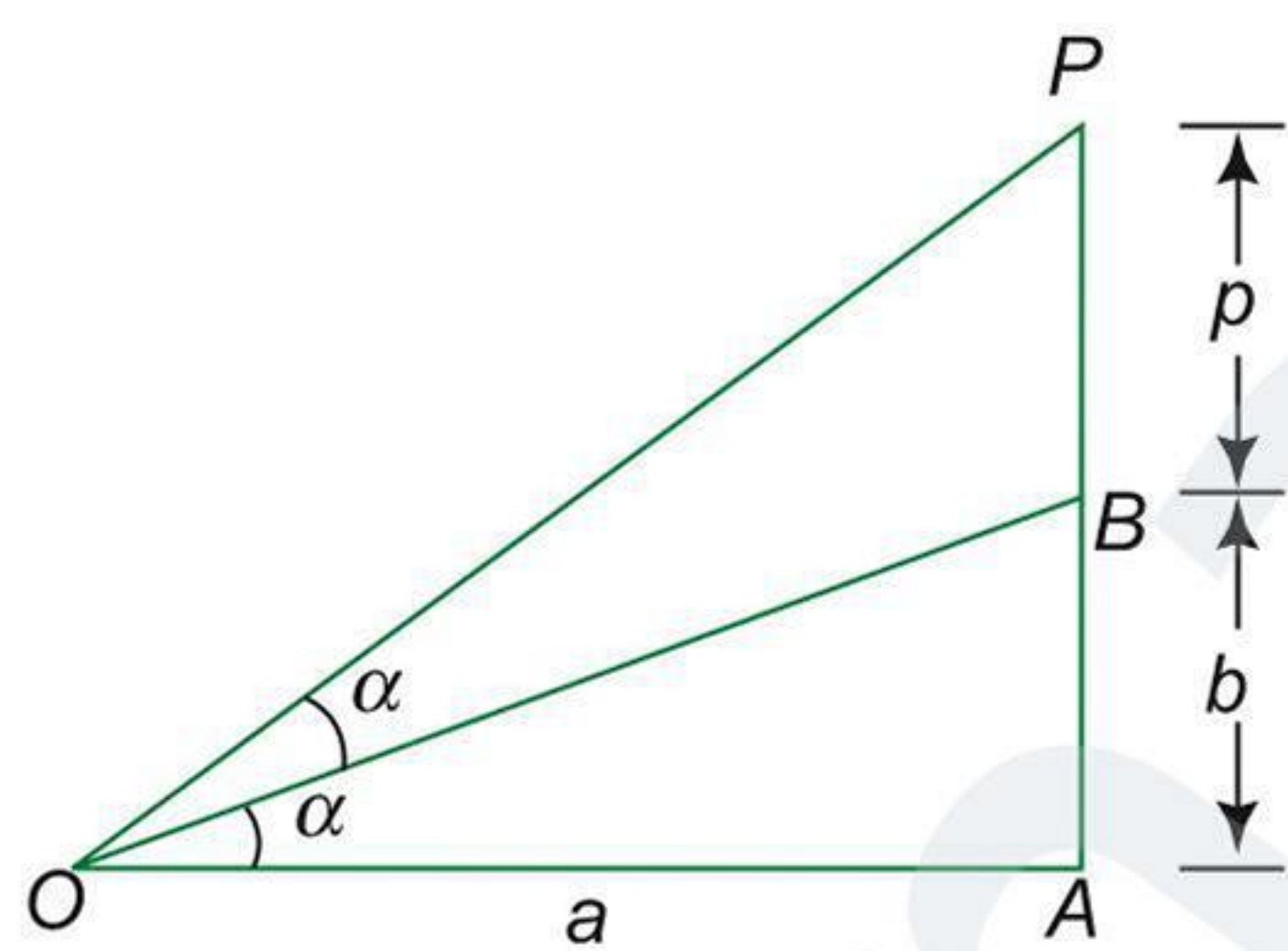
$$\therefore h = 120 \text{ m}$$

$$\Rightarrow 120 + x = 120\sqrt{3}$$

$$\Rightarrow x = 120(\sqrt{3} - 1) \text{ m}$$



3. (4) Let AB be the tower and PB be the pole.



From the figure, $\tan \alpha = \frac{b}{a}$

$$\text{and } \tan 2\alpha = \frac{p+b}{a}$$

$$\therefore \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{p+b}{a}$$

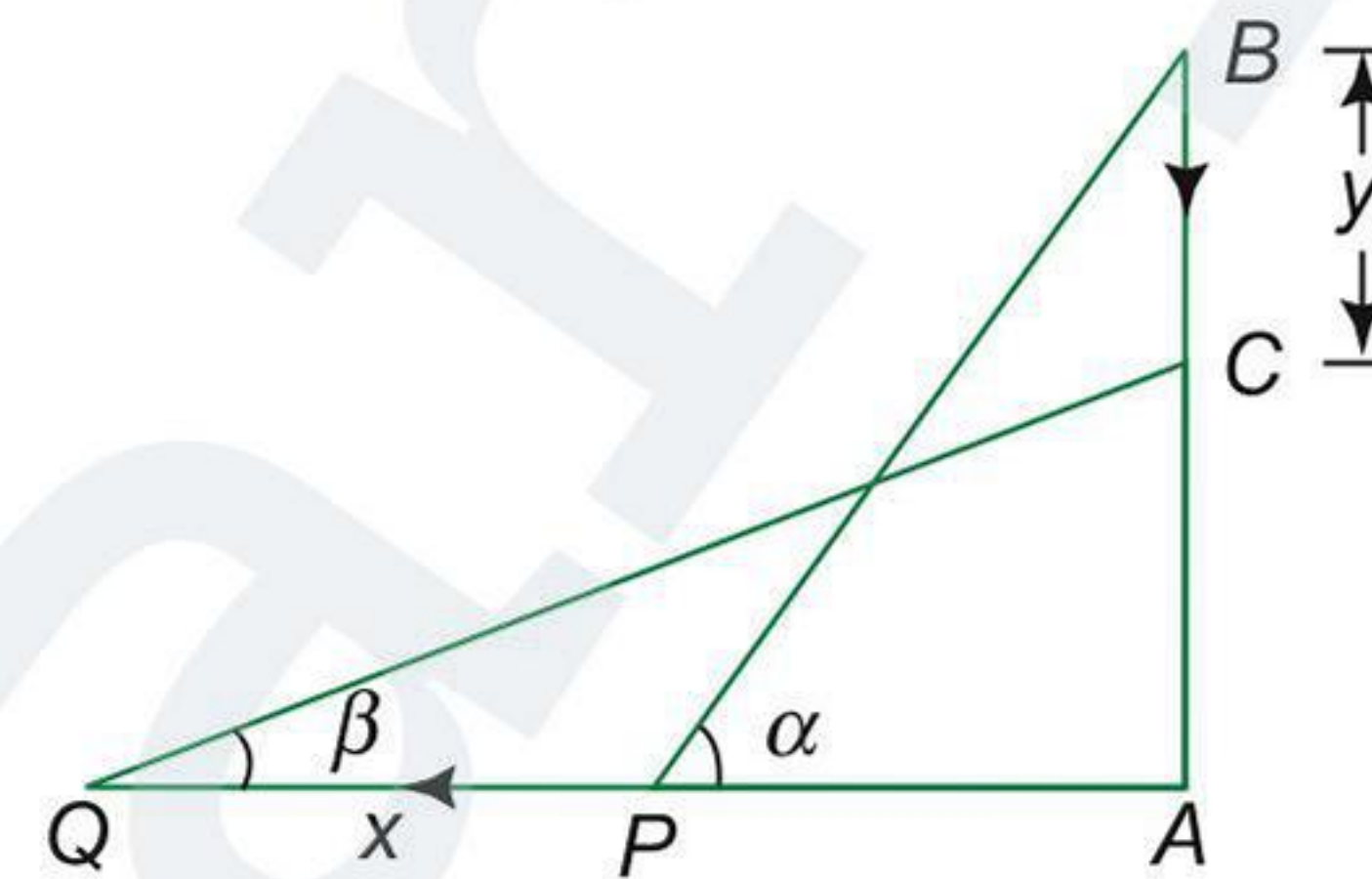
$$\Rightarrow \frac{2\left(\frac{b}{a}\right)}{1 - \left(\frac{b}{a}\right)^2} = \frac{p+b}{a}$$

$$\Rightarrow \frac{2ba}{a^2 - b^2} = \frac{p+b}{a}$$

$$\Rightarrow \frac{2ba^2 - a^2b + b^3}{a^2 - b^2} = p$$

$$\Rightarrow p = b \left(\frac{a^2 + b^2}{a^2 - b^2} \right)$$

4. (2) Let AB be the wall, P be the initial position and Q be the final position of the foot of the ladder on the ground.



In the figure, $PB = QC = l$

(Length of ladder)

In triangle BAP , $PA = l \cos \alpha$, $AB = l \sin \alpha$

In triangle CAQ , $QA = l \cos \beta$, $AC = l \sin \beta$

Now, $y = BC = AB - AC = l(\sin \alpha - \sin \beta)$

and $x = PQ = AQ - AP = l(\cos \beta - \cos \alpha)$

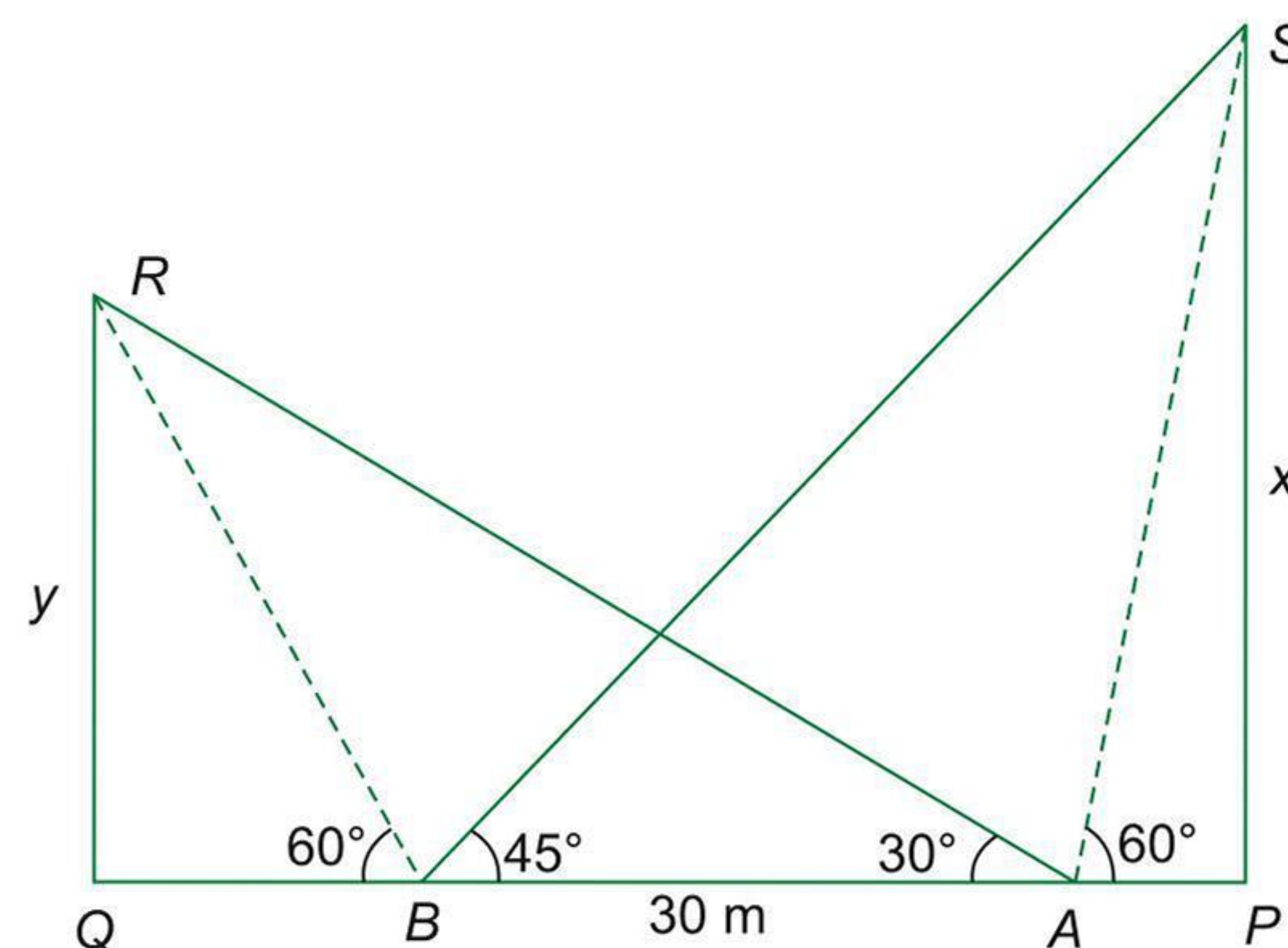
$$\Rightarrow \frac{y}{x} = \frac{\sin \alpha - \sin \beta}{\cos \beta - \cos \alpha}$$

$$= \frac{2 \sin \left(\frac{\alpha - \beta}{2} \right) \cos \left(\frac{\alpha + \beta}{2} \right)}{2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)}$$

$$\Rightarrow \frac{y}{x} = \cot \left(\frac{\alpha + \beta}{2} \right)$$

$$\Rightarrow x = y \tan \left(\frac{\alpha + \beta}{2} \right)$$

5. (4) Let x and y be the heights of the flagstaffs at P and Q , respectively.



$$\text{Then } AP = x \cot 60^\circ = \frac{x}{\sqrt{3}},$$

$$AQ = y \cot 30^\circ = y\sqrt{3},$$

$$BP = x \cot 45^\circ = x,$$

$$\text{and } BQ = y \cot 60^\circ = \frac{y}{\sqrt{3}}$$

$$\therefore AB = BP - AP = x - \frac{x}{\sqrt{3}}$$

$$\Rightarrow 30\sqrt{3} = (\sqrt{3} - 1)x$$

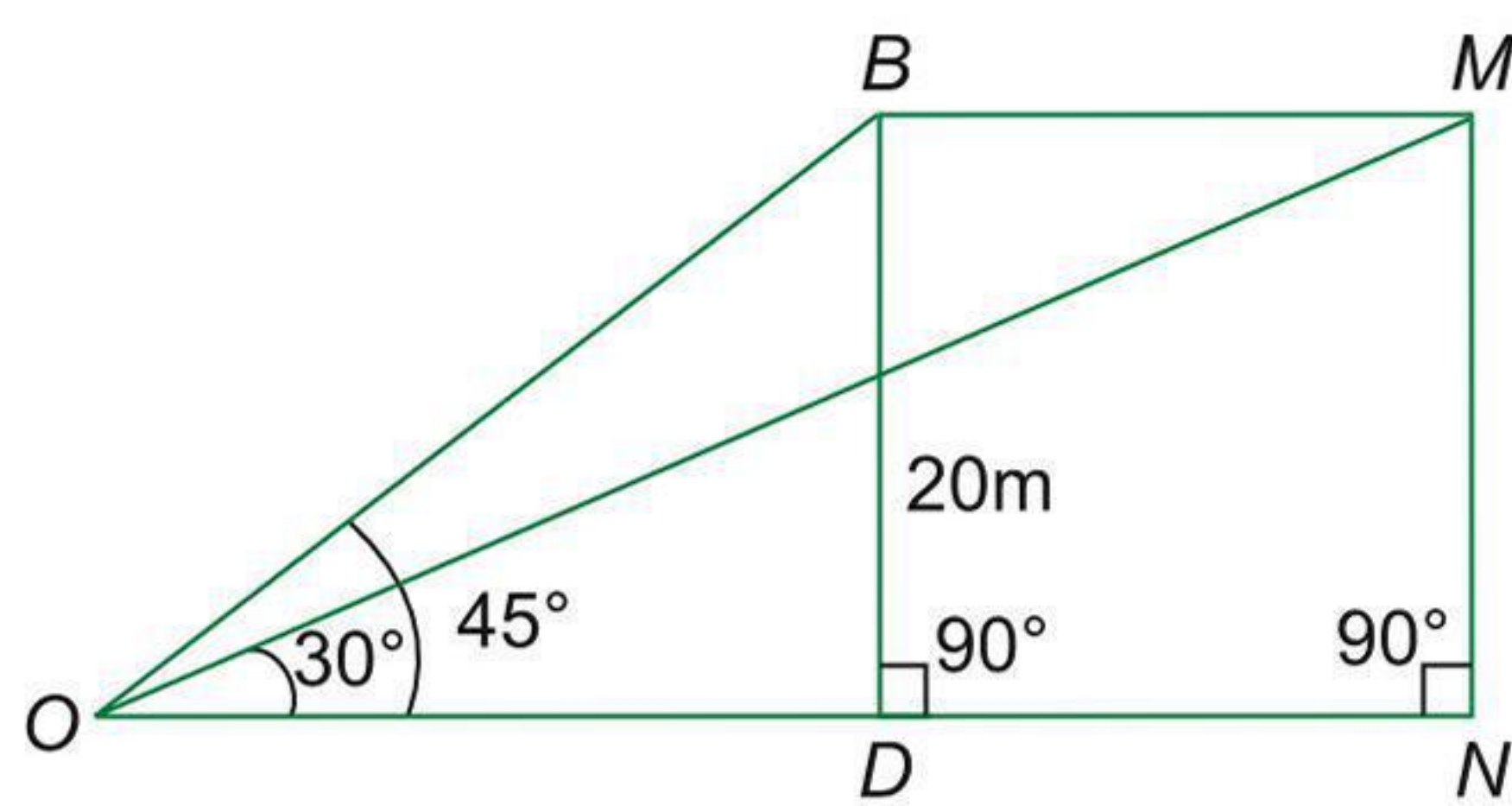
$$\Rightarrow x = 15(3 + \sqrt{3})$$

$$\text{Similarly, } 30 = y\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)$$

$$\text{or } y = 15\sqrt{3}$$

$$\begin{aligned}\therefore PQ &= BP + BQ = x + \frac{y}{\sqrt{3}} \\ &= 15(3 + \sqrt{3}) + 15 \\ &= (60 + 15\sqrt{3}) \text{ m}\end{aligned}$$

6. (1) Let DB be the pole and the snake be at point O .
Then $\angle BOD = 45^\circ$ and $BD = 20$ metres.
Now, the eagle flies horizontally and reaches at point M in 1 second.



Then $\angle MON = 30^\circ$, where $MN \perp ON$

Now, $BD = MN = 20$ metres

From triangle BOD , $OD = 20$ metres

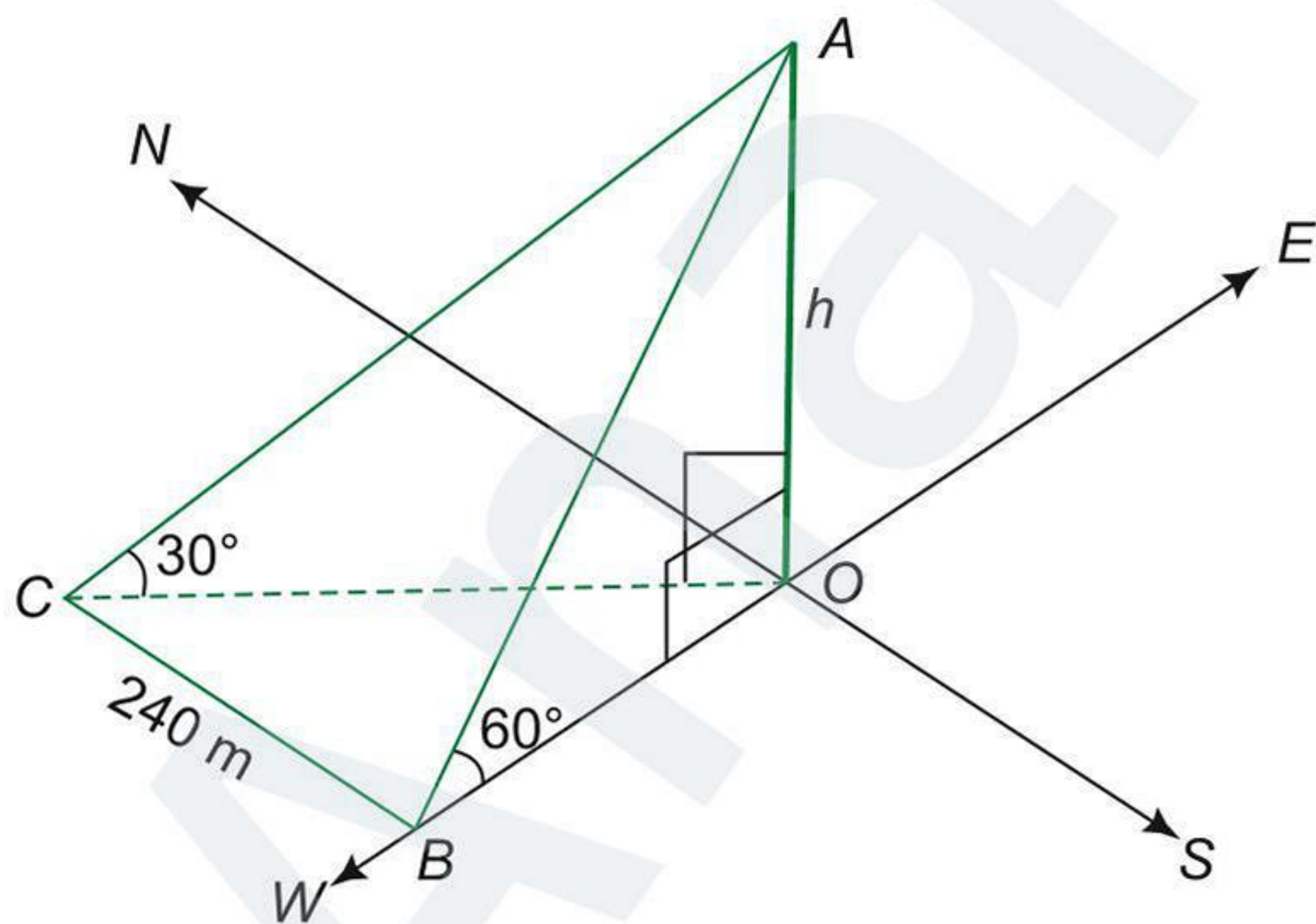
Again from $\triangle MON$,

$$\tan 30^\circ = \frac{MN}{ON} = \frac{20}{20 + DN}$$

$$\therefore DN = 20(\sqrt{3} - 1) = 20 \times 0.732 = 14.64 \text{ metres}$$

$$\therefore \text{Speed of eagle} = \frac{\text{Distance}}{\text{time}} = \frac{BM}{1} = \frac{DN}{1} = 14.64 \text{ m/s}$$

7. (3) Let OA be the tower of height h . Also, let B and C be the initial and final positions, respectively, of the man.



$$\text{In triangle } AOB, OB = h \cot 60^\circ = \frac{h}{\sqrt{3}}.$$

$$\text{In triangle } AOC, OC = h \cot 30^\circ = \sqrt{3}h.$$

Now in triangle OBC ,

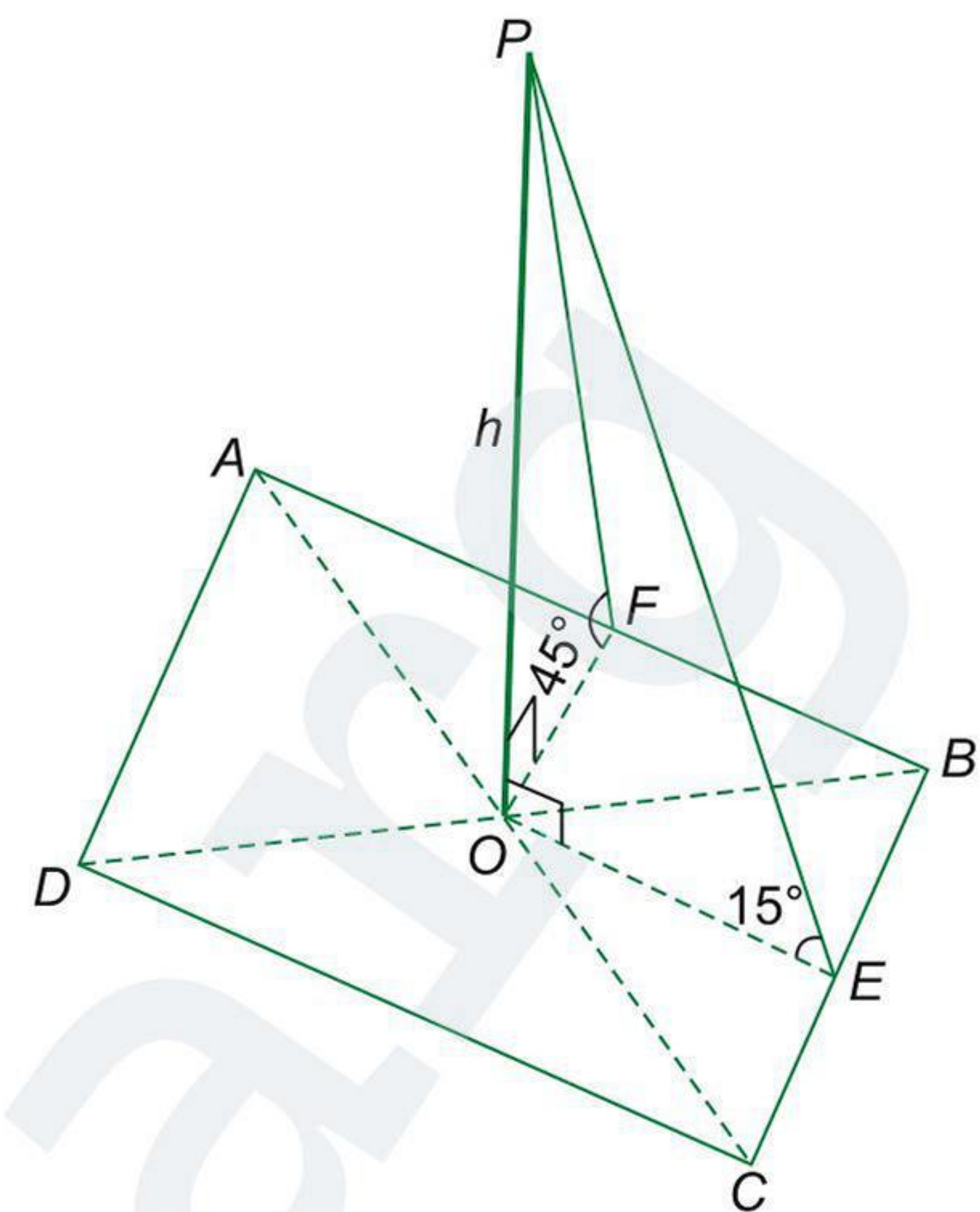
$$OB^2 + BC^2 = OC^2$$

$$\therefore \frac{h^2}{3} + (240)^2 = 3h^2$$

$$\Rightarrow \frac{8h^2}{3} = (240)^2$$

$$\Rightarrow h = 60\sqrt{6} \text{ m}$$

8. (3) Let OP be the flagstaff of height ' h ' standing at the centre O of the rectangular field $ABCD$ subtending angles 15° and 45° at E and F , respectively.



In triangle POE , $OE = h \cot 15^\circ = h(2 + \sqrt{3})$

In triangle POF , $OF = h \cot 45^\circ = h$

$$\begin{aligned}\Rightarrow EF &= \sqrt{h^2 + (2 + \sqrt{3})^2 h^2} \\ &= 2h\sqrt{2 + \sqrt{3}}\end{aligned}$$

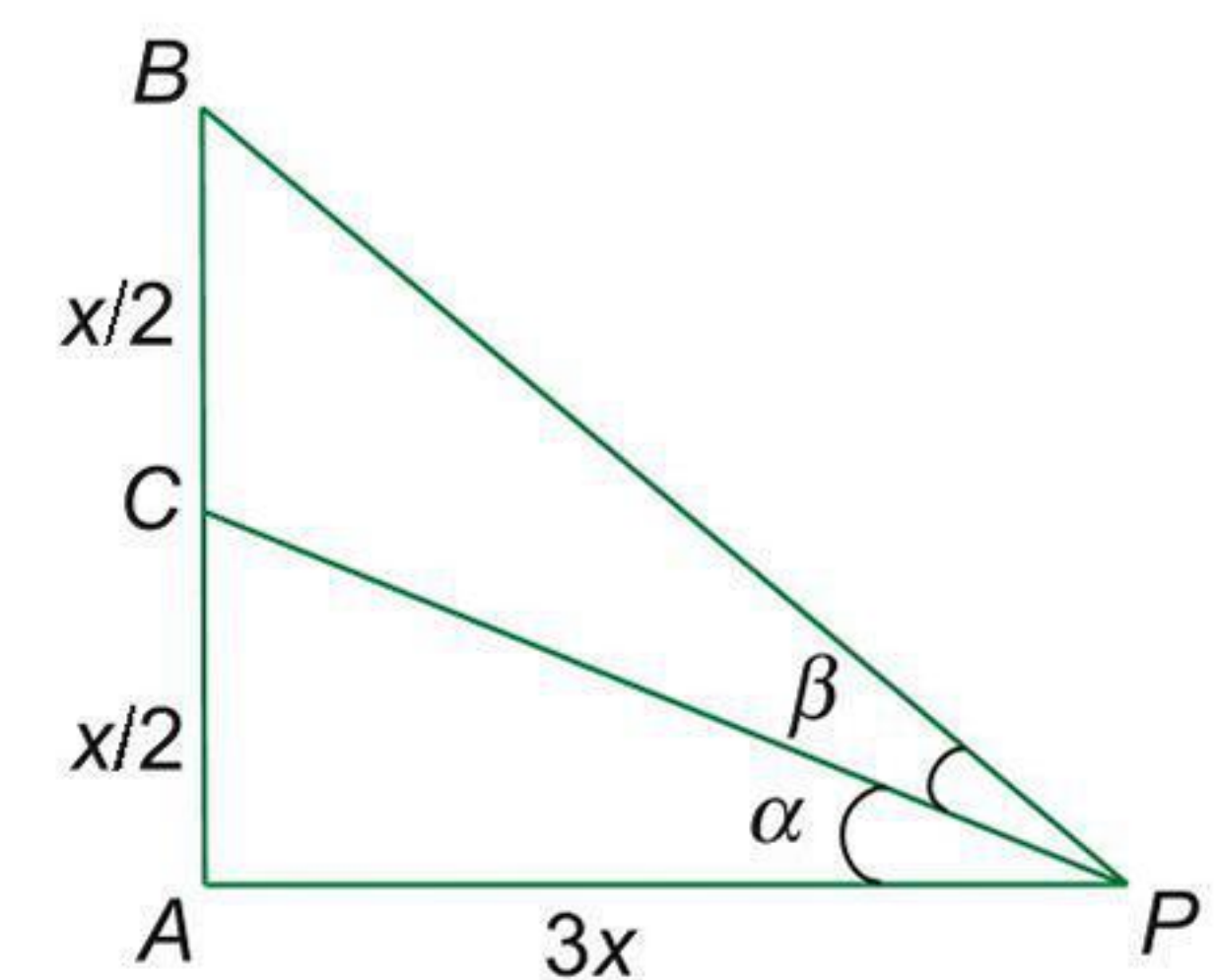
Triangles ACB and FEB are similar.

So, $AC = 2EF$

$$\Rightarrow 120 = 4h\sqrt{2 + \sqrt{3}}$$

$$\Rightarrow h = \frac{30}{\sqrt{2 + \sqrt{3}}} = 30\sqrt{2 - \sqrt{3}}$$

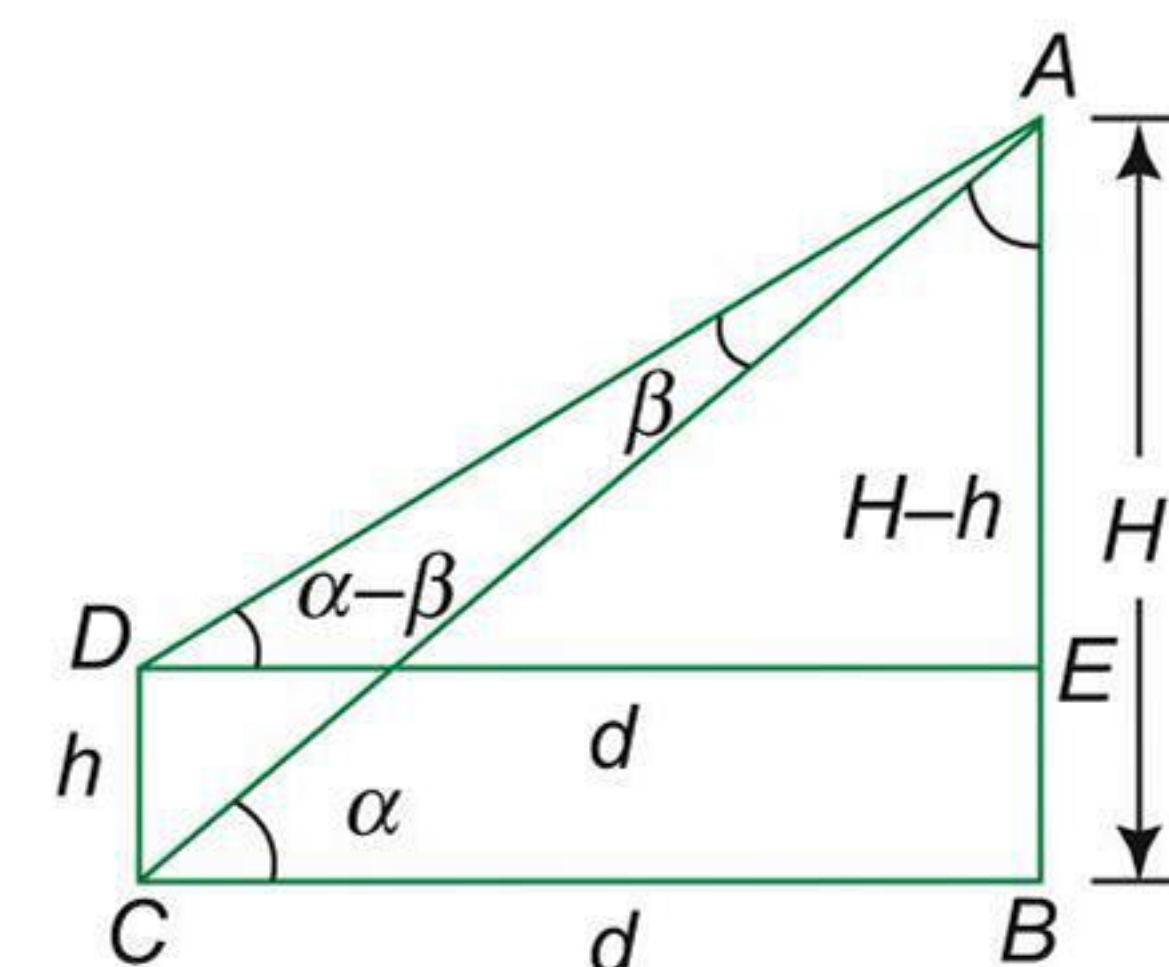
9. (2) On the basis of the given information, following diagram is obtained.



$$\tan \beta = \tan ((\alpha + \beta) - \alpha)$$

$$= \frac{\frac{1}{3} - \frac{1}{6}}{1 + \frac{1}{3} \cdot \frac{1}{6}} = \frac{3}{19}$$

10. (1) Let AB be the tower of height H and CD be the pole.



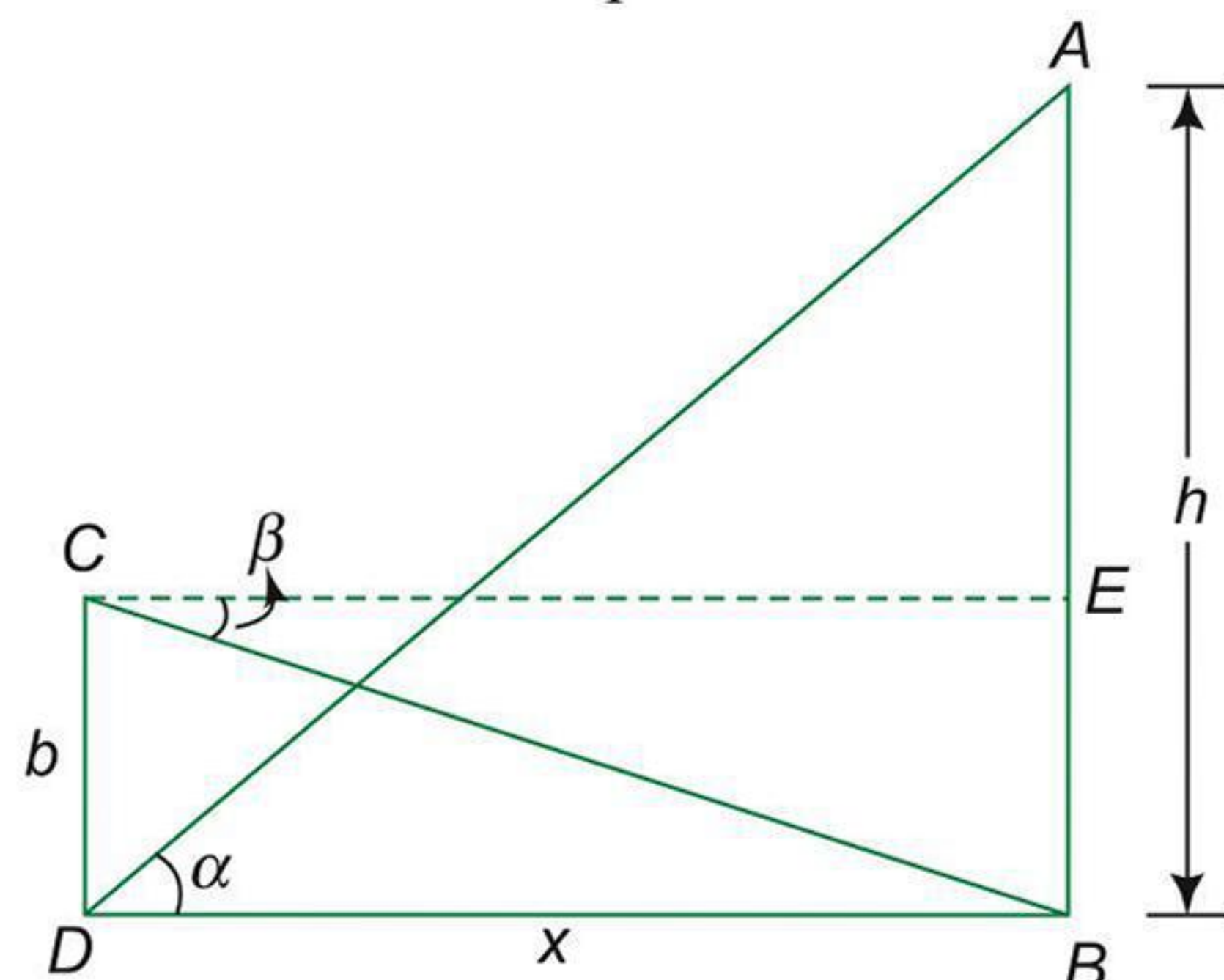
In triangle ABC , $d = H \cot \alpha$

In triangle AED , $d = (H - h) \cot (\alpha - \beta)$

$$\Rightarrow H \cot \alpha = (H - h) \cot (\alpha - \beta)$$

$$\therefore H = \frac{h \cot(\alpha - \beta)}{\cot(\alpha - \beta) - \cot \alpha}$$

11. (3) Let AB be the tower of height h , D be the point x m away from B and C be another point b m above D .



In triangle ABD ,

$$h = x \tan \alpha \quad \dots (1)$$

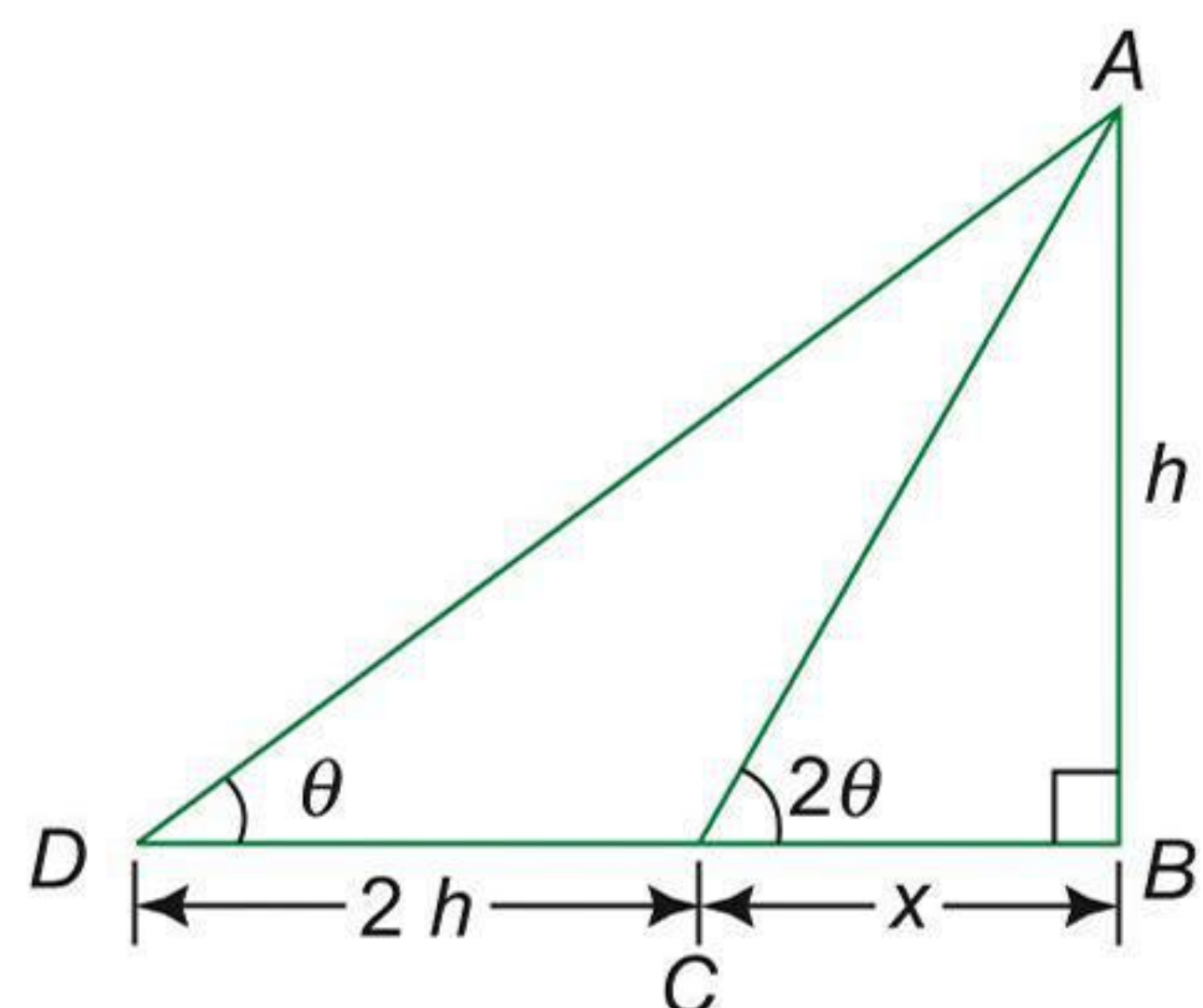
In triangle BCE ,

$$x = b \cot \beta \quad \dots (2)$$

From (1) and (2), we get

$$\therefore h = b \tan \alpha \cot \beta$$

12. (4) Let AB be the pole of height h . Also, let D and C be the initial and final positions of the man.



In triangle ABC ,

$$x = h \cot 2\theta \quad \dots (1)$$

In triangle ABD ,

$$h = (2h + x) \tan \theta \quad \dots (2)$$

Put the value of x from (1) into (2), we get

$$1 = \left(2 + \frac{1}{\tan 2\theta}\right) \tan \theta$$

$$\Rightarrow 1 = \left(2 + \frac{1 - \tan^2 \theta}{2 \tan \theta}\right) \tan \theta$$

$$\therefore \tan^2 \theta - 4 \tan \theta + 1 = 0$$

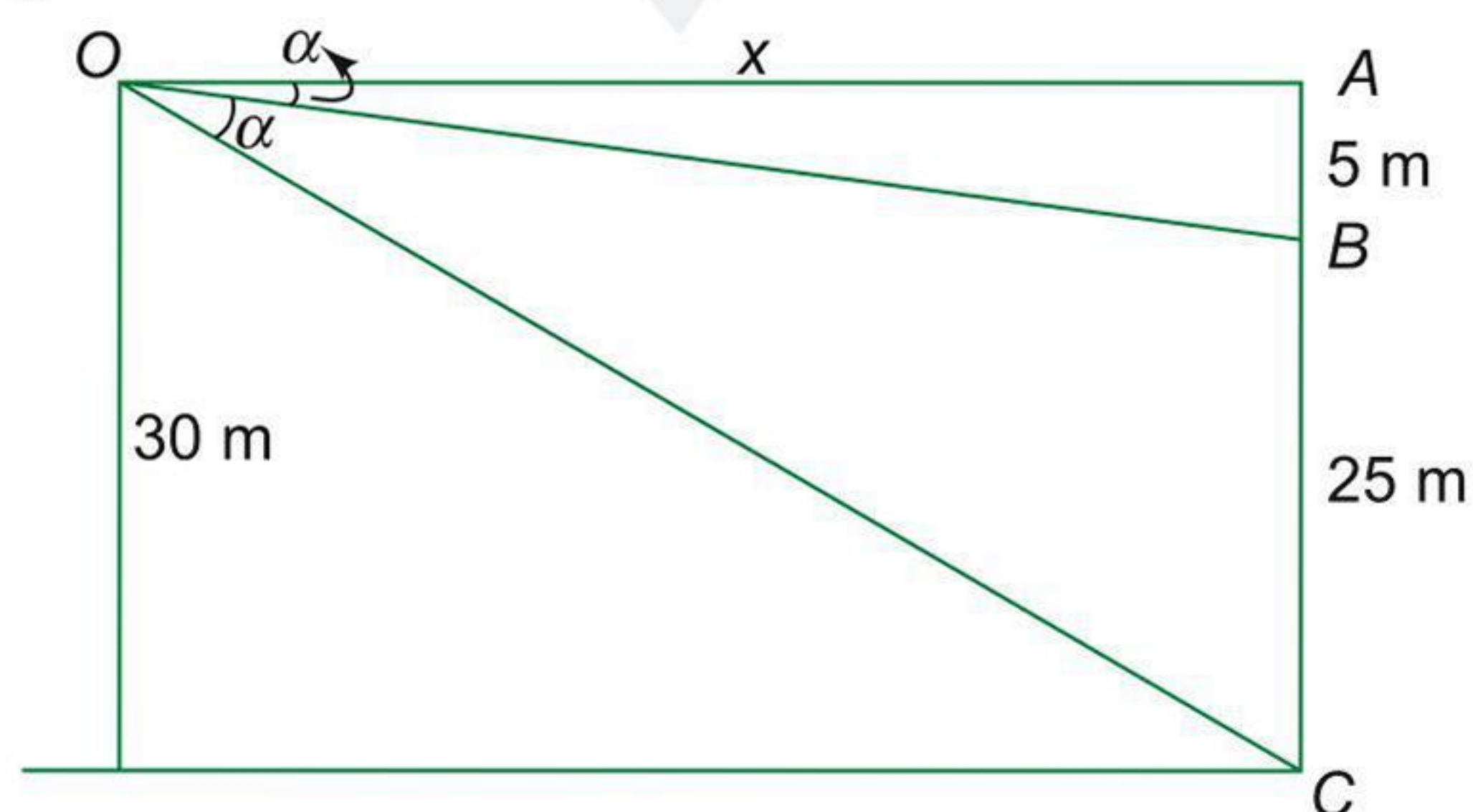
$$\Rightarrow \tan \theta = 2 \pm \sqrt{3}$$

$$\tan \theta = 2 + \sqrt{3} \quad (\text{Rejected as otherwise } 2\theta = 150^\circ)$$

$$\therefore \tan \theta = (2 - \sqrt{3})$$

$$\Rightarrow \theta = 15^\circ$$

13. (3) Let AB be the pole standing on the building BC and O be the position of the antenna.



In triangle OAB ,

$$\tan \alpha = \frac{5}{x} \quad \dots (1)$$

In triangle OAC ,

$$\tan 2\alpha = \frac{30}{x} \quad \dots (2)$$

From (1) and (2),

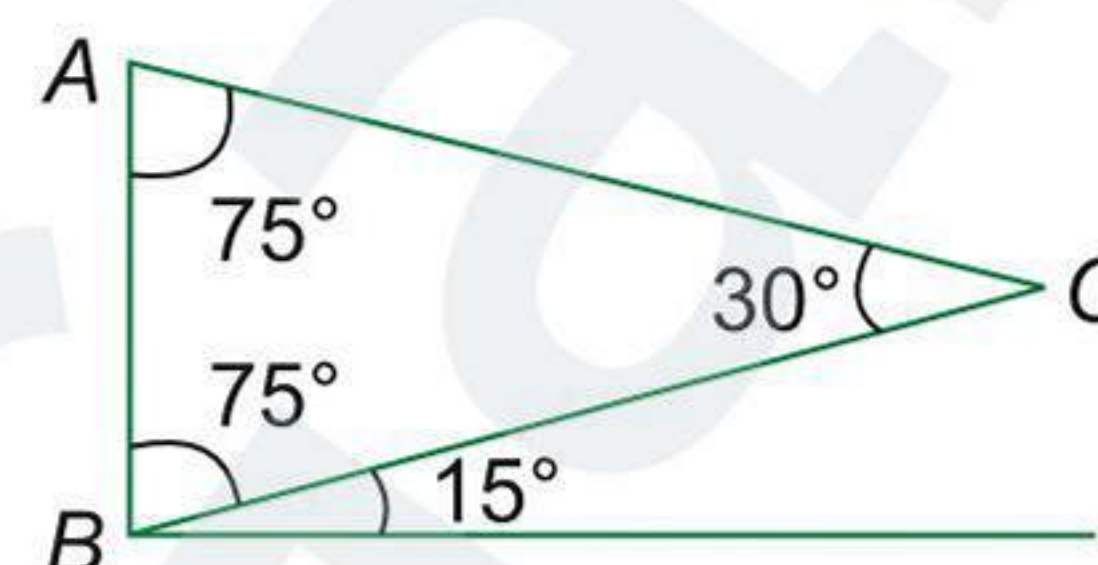
$$\tan 2\alpha = 6 \tan \alpha$$

$$\Rightarrow 3 - 3 \tan^2 \alpha = 1$$

$$\Rightarrow \tan \alpha = \sqrt{\frac{2}{3}}$$

$$\text{From (1), } x = 5 \cot \alpha = 5\sqrt{\frac{3}{2}}$$

14. (3) Let BC be the declivity and BA be the tower.



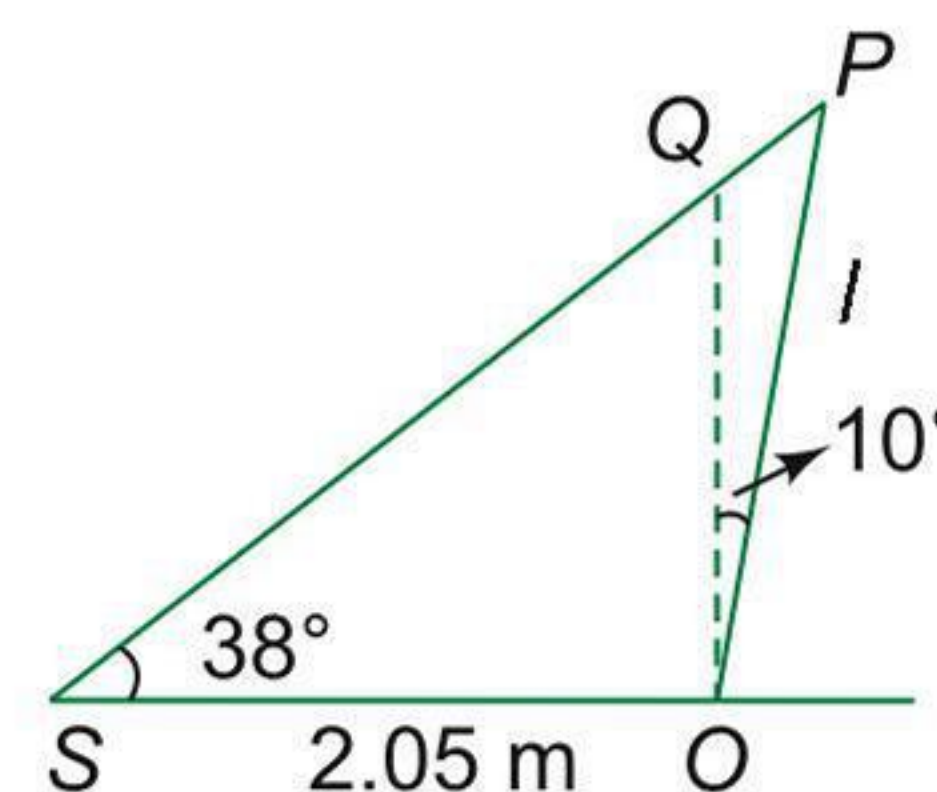
Using sine rule in $\triangle ABC$, we get

$$\frac{BC}{\sin 75^\circ} = \frac{AB}{\sin 30^\circ}$$

$$\Rightarrow AB = \frac{80 \sin 30^\circ}{\sin 75^\circ}$$

$$\Rightarrow AB = \frac{40 \times 2\sqrt{2}}{\sqrt{3} + 1} = 40(\sqrt{6} - \sqrt{2})$$

15. (3) Let PO be the tower and OS be its shadow.



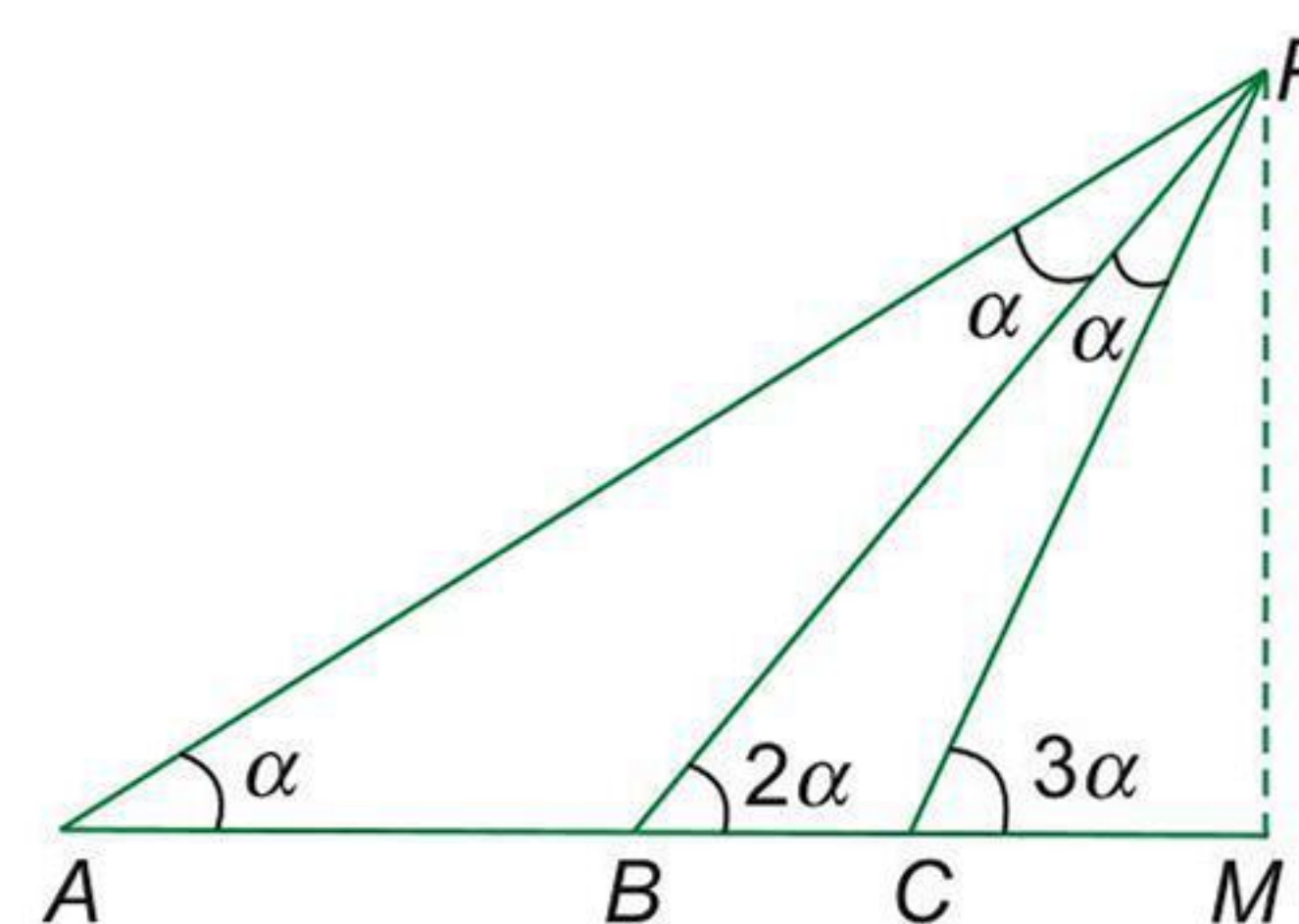
In triangle SOP , using sine rule,

$$\frac{l}{\sin 38^\circ} = \frac{2.05}{\sin(\angle SPO)} \quad \dots (1)$$

$$\angle SPO = 180^\circ - 38^\circ - 90^\circ - 10^\circ = 42^\circ$$

$$\text{So, from (1), } l = \frac{2.05 \sin 38^\circ}{\sin 42^\circ}$$

16. (2) Let PM be the tower.



Given $\angle BAP = \alpha$, $\angle CBP = 2\alpha$ and $\angle MCP = 3\alpha$

$$\therefore \angle APB = \angle PBC - \angle BAP = 2\alpha - \alpha = \alpha$$

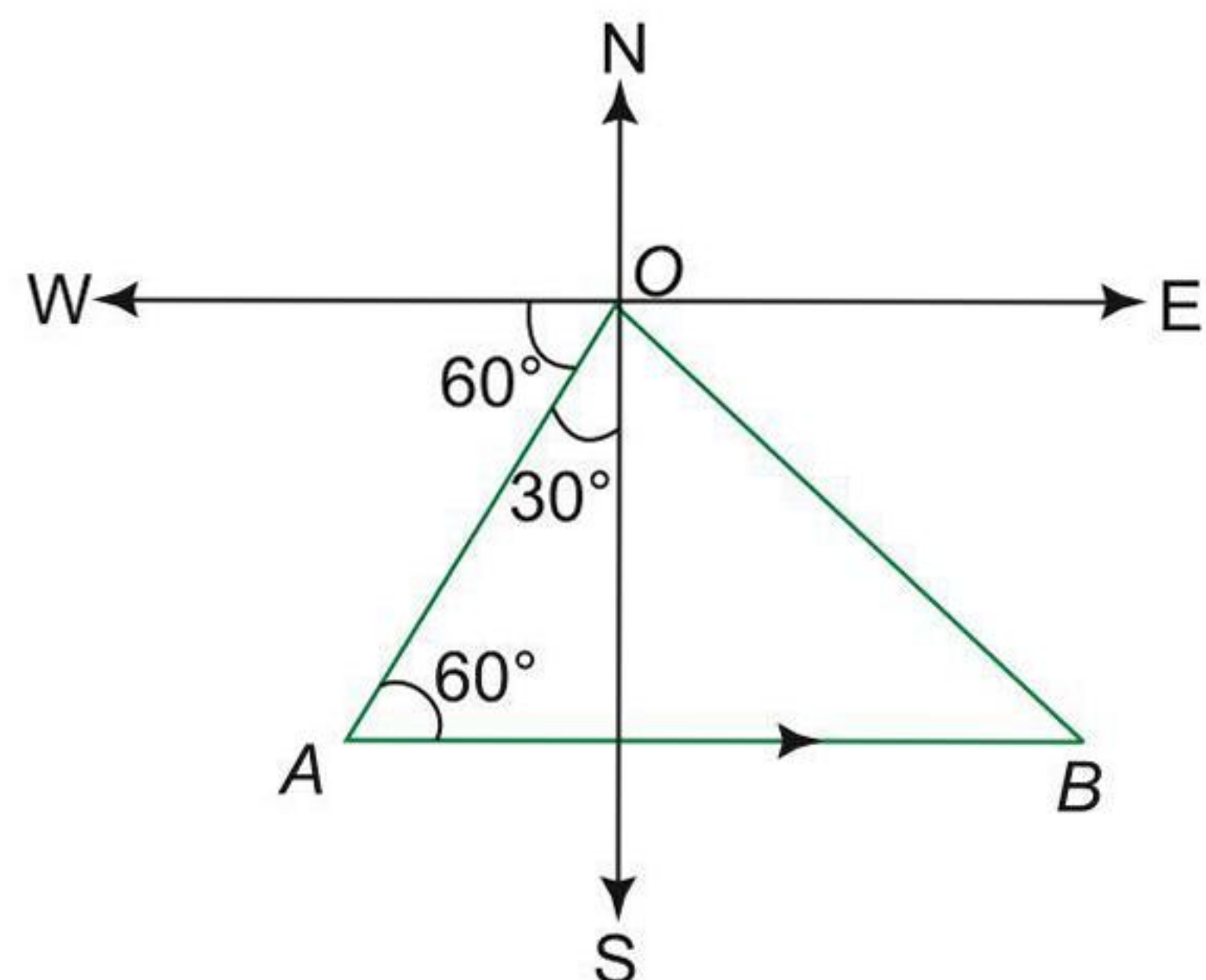
Similarly, $\angle BPC = \alpha$

$$\Rightarrow BP = AB$$

By sine formula in triangle BCP , we have

$$\begin{aligned}
 \frac{BC}{\sin(\angle BPC)} &= \frac{BP}{\sin(\angle BCP)} \\
 \Rightarrow \frac{BC}{\sin \alpha} &= \frac{AB}{\sin(180^\circ - 3\alpha)} \\
 \Rightarrow \frac{AB}{BC} &= \frac{\sin 3\alpha}{\sin \alpha} = \frac{3 \sin \alpha - 4 \sin^3 \alpha}{\sin \alpha} \\
 &= 3 - 4 \sin^2 \alpha \\
 &= 3 - 2(1 - \cos 2\alpha) \\
 &= 1 + 2 \cos 2\alpha
 \end{aligned}$$

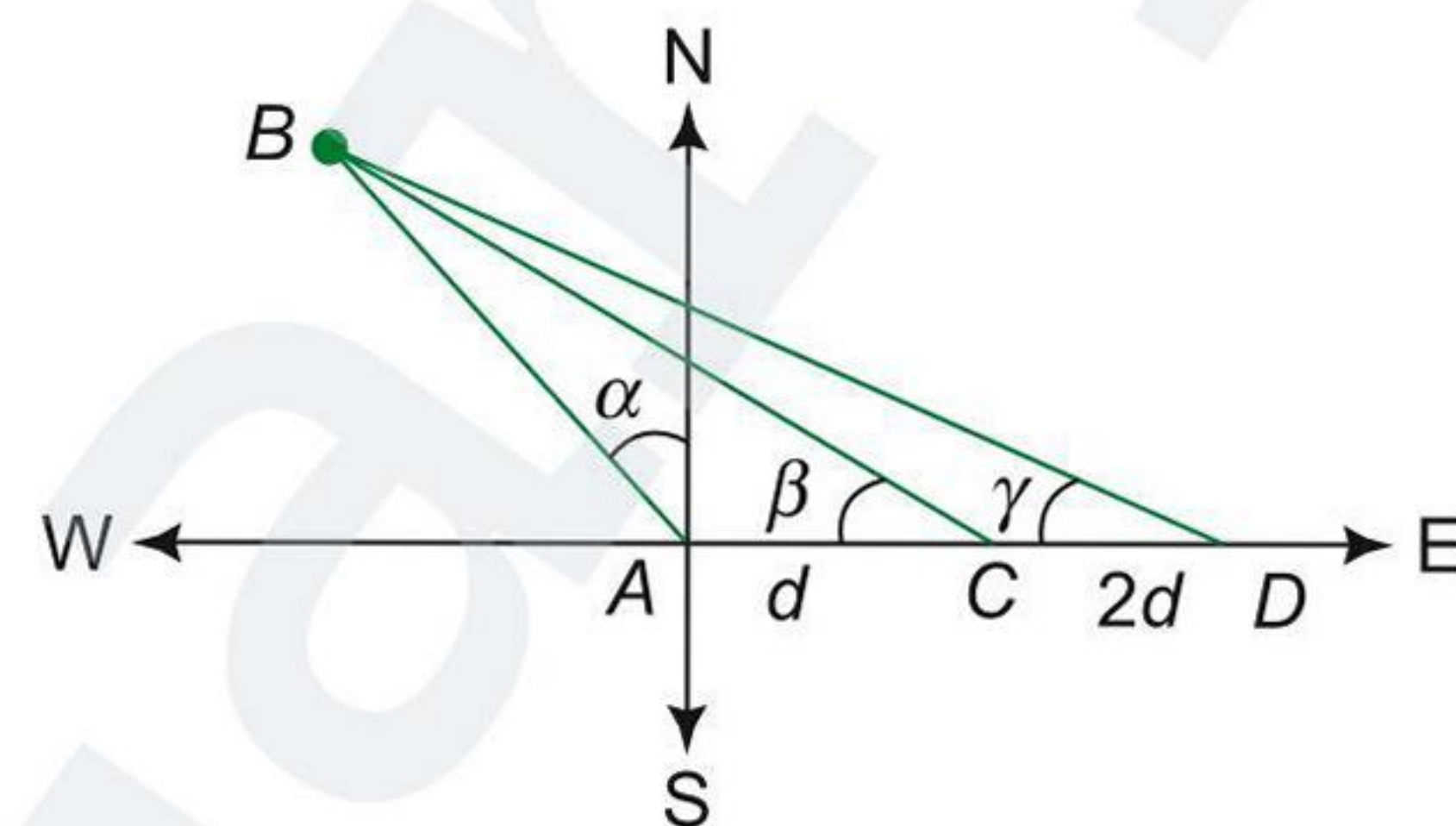
17. (2) Let A be the position of the harbour and O be the fort. $OA = 30$, $OB = 70$. Let $AB = x$.



Applying cosine formula in $\triangle OAB$, we get

$$\begin{aligned}
 \cos 60^\circ &= \frac{OA^2 + AB^2 - OB^2}{2OA \cdot AB} \\
 \Rightarrow \frac{1}{2} &= \frac{900 + x^2 - 4900}{2 \times 30 \times x} \\
 \Rightarrow x^2 - 30x - 4000 &= 0 \\
 \Rightarrow x &= 80 \\
 AB &= 80 \text{ km} \\
 \text{Speed of ship} &= 10 \text{ km/hr} \\
 \therefore \text{Time} &= 8 \text{ hr}
 \end{aligned}$$

18. (2) Observe the following figure.



Using $m-n$ theorem in triangle ABD , we have

$$\begin{aligned}
 (d + 2d) \cot \beta &= d \cot \gamma - 2d \cot (90^\circ + \alpha) \\
 \Rightarrow 3d \cot \beta &= d \cot \gamma + 2d \tan \alpha \\
 \Rightarrow 3 \cot \beta &= \cot \gamma + 2 \tan \alpha \\
 \therefore 2 \tan \alpha &= 3 \cot \beta - \cot \gamma
 \end{aligned}$$